

2. Requirement Analysis

The requirement analysis phase involved identifying the functional and non-functional requirements necessary for implementing the Smart Laptop Request Management system. Discussions were conducted to understand the expectations of employees, managers, and IT administrators who would interact with the application.

Functional Requirements:

From a functional perspective, employees must be able to independently submit a laptop request by providing a standard set of attributes through an accessible interface. The input profile must capture data points including employee name, corporate ID, home department, project alignment type, job role, and an explicit business justification statement. To prevent user confusion and reduce typographical errors, user select from predefined equipment choices rather than manually typing computer configurations. Once a request passes initial validation, the system must trigger automated approval routing to the appropriate line manager without human intervention.

Managers require a simplified interactive system interface to execute approvals or rejections directly from emails or portal dashboards. Upon approval, the system must automatically generate an active catalog task routed directly to the IT provisioning desk queue. Conversely, if a manager rejects a request, the system must immediately change the ticket status to Closed Rejected, stop downstream workflows, and alert the employee. Throughout this lifecycle, the requester must have on-demand access to real-time status updates via their portal dashboard, keeping them informed at every stage of the provisioning process.

Non-Functional Requirements :

Non-functional requirements were equally crucial to ensuring long-term operational success, platform stability, and high user adoption. The interface layer must be fully responsive and integrated into the existing ServiceNow Service Portal framework, presenting a consistent corporate brand experience. Security mechanisms must enforce strict role-based access controls, ensuring that only authenticated, active employees can view the form and submit requests, while preventing unauthorized external script submittals.

The system's underlying processing logic must be structured to scale, handling multiple simultaneous request submissions during corporate hiring surges or quarterly hardware refreshes without causing thread lockups or portal lag. High availability and continuous uptime are baseline mandates, as employees working across global regions may need to submit hardware requests outside of standard local business hours. Data retention and audit tracking policies must be enforced at the database level to maintain records for compliance verification cycles.

System Operational Scope Boundaries Matrix:

Requirement Area	Target Threshold Specification	Verification Metric
Form Load Performance	Render complete catalog item under 1.8 seconds.	Automated ATF Test Runner Script
Concurrent Transaction Capacity	Process up to 250 parallel submissions simultaneously.	Load Testing Performance Profile
Access Controls & Security	Enforce User-Criteria limits linked to active LDAP profiles.	Security Role Matrix Audit Log
System Operational Uptime	99.9% availability for 24/7 global submission access.	ServiceNow Cloud SLA Metrics

By mapping out these distinct requirement blocks, the implementation team created a solid functional roadmap. This rigorous definition prevented scope creep, allowed engineers to map inputs directly to specific database fields, and formed the baseline for all subsequent automated automated testing procedures.

Catalog Item

Request Laptop

Copy

Try It

Update

Edit in Catalog Builder

Delete

Related Links

[Item Diagnostic](#)

[Run Point Scan](#)

Variables (8)

Variable Sets

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

Order

Search

⌕

Actions on selected rows...

New

Catalog item = Request Laptop

Type	Question	Order
Multi Line Text	Business Justification	
Single Line Text	Employee Name	
Select Box	Laptop Type	
Single Line Text	Employee ID	
Select Box	Project Type	
Select Box	Request Type	
Select Box	Job Role	
Select Box	Department	

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